

Assignment 03

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CSE260: Digital Logic Design

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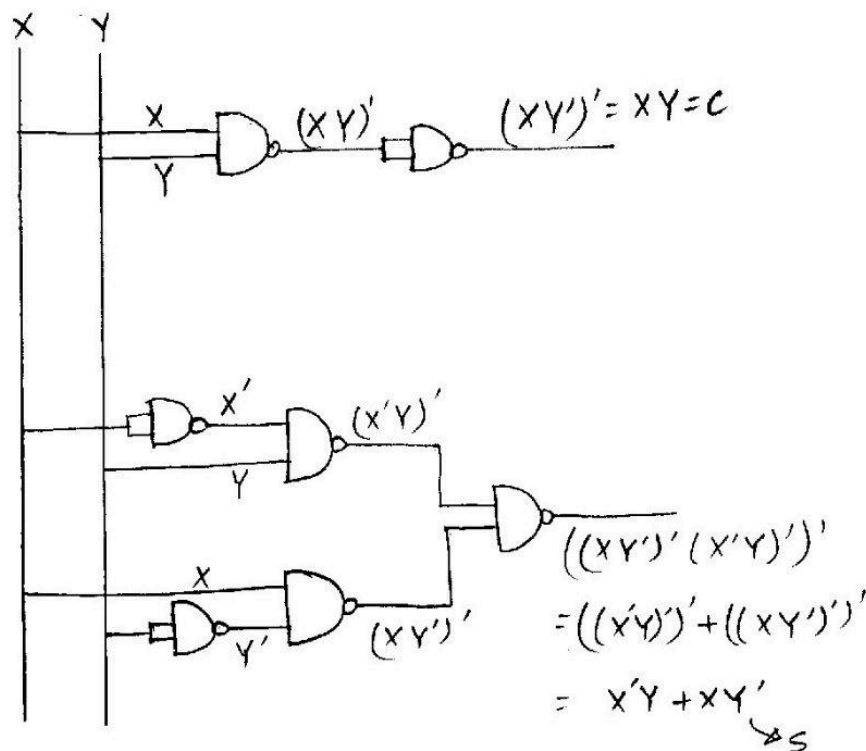
April 24, 2025

Ans to the ques no-01

To design a half adder using NAND gate only.

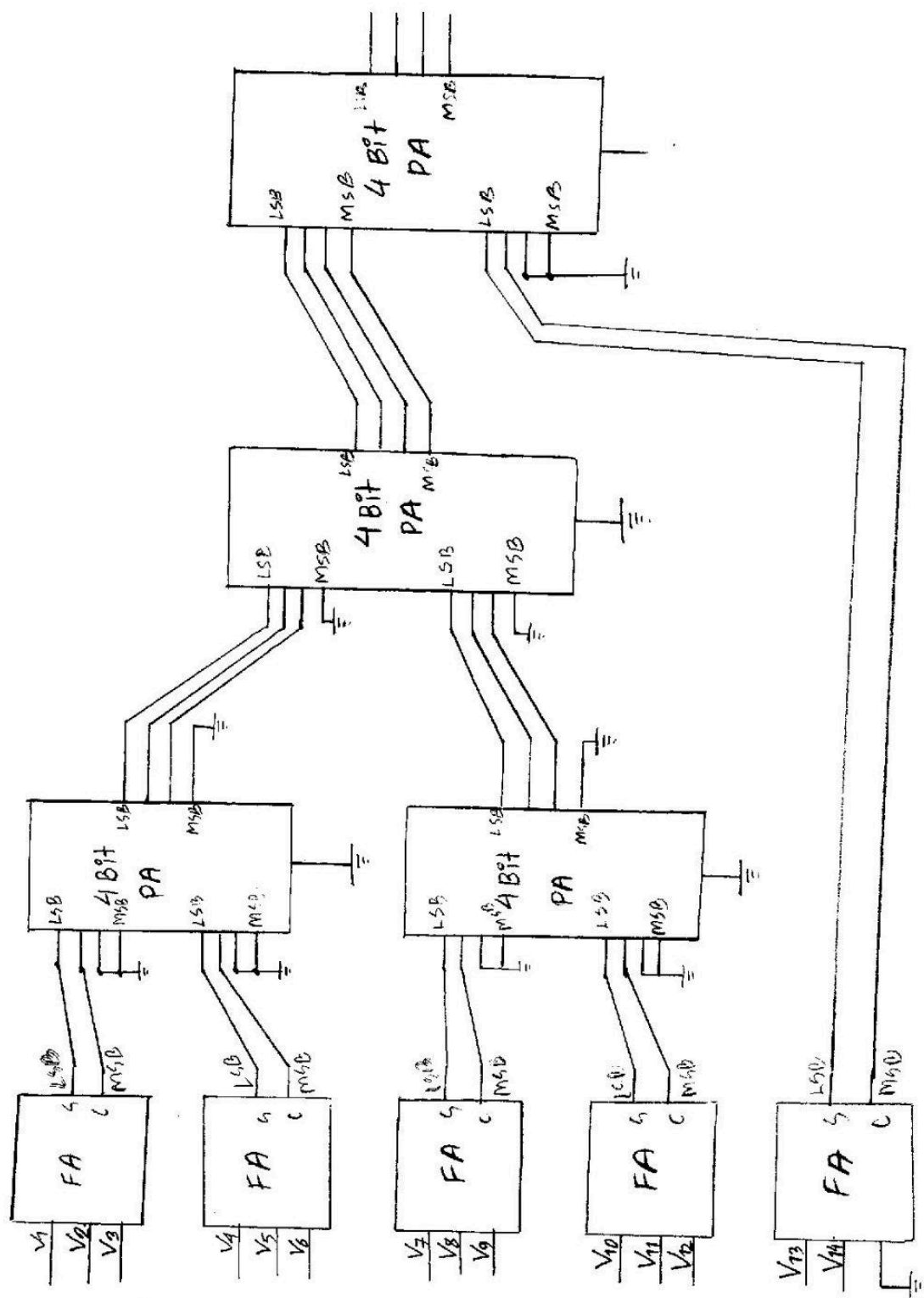
For half adder,

carry bit, $C = XY$
sum bit, $S = X'Y + XY'$ } where X and Y are the two bit digits



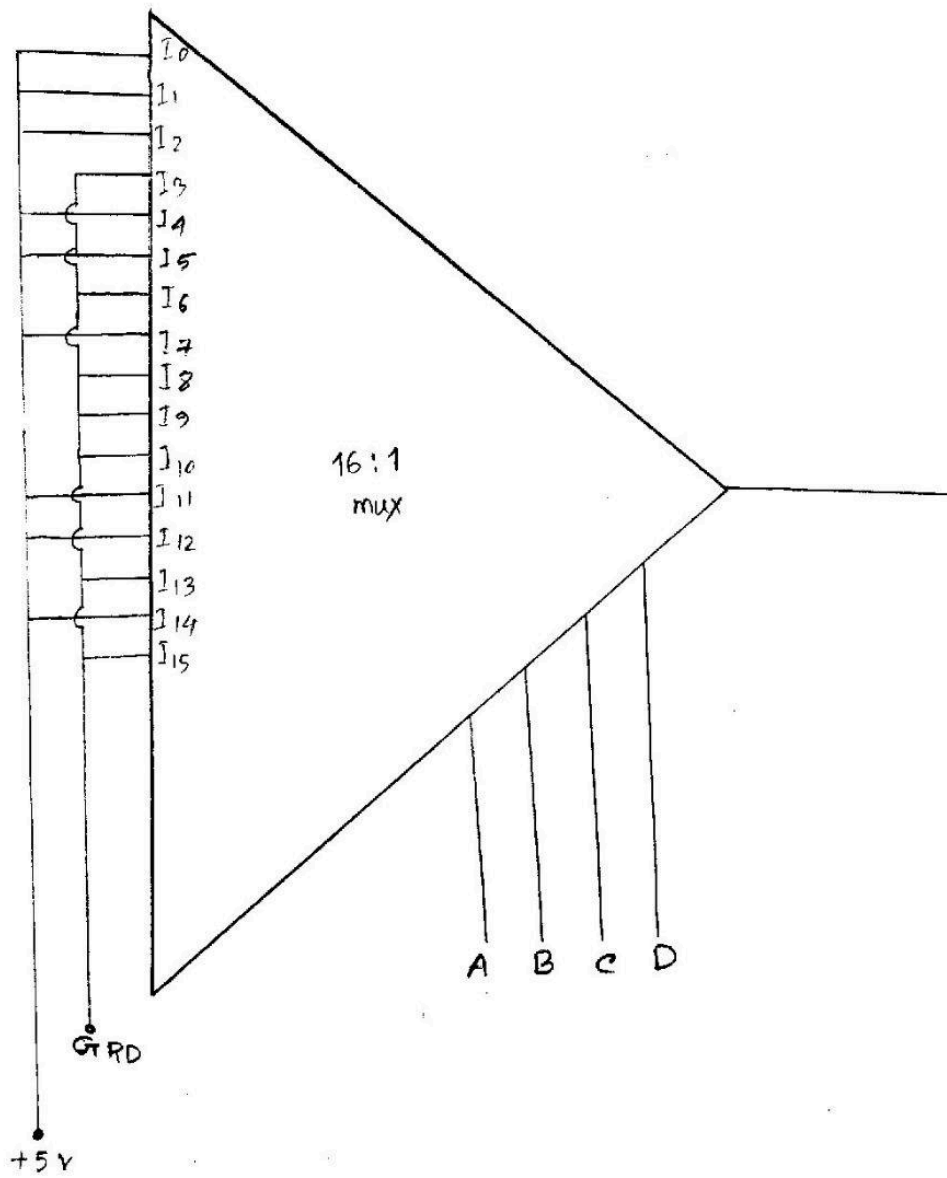
Ans to the ques no-03

14 person voting system
using full adders and parallel adders.

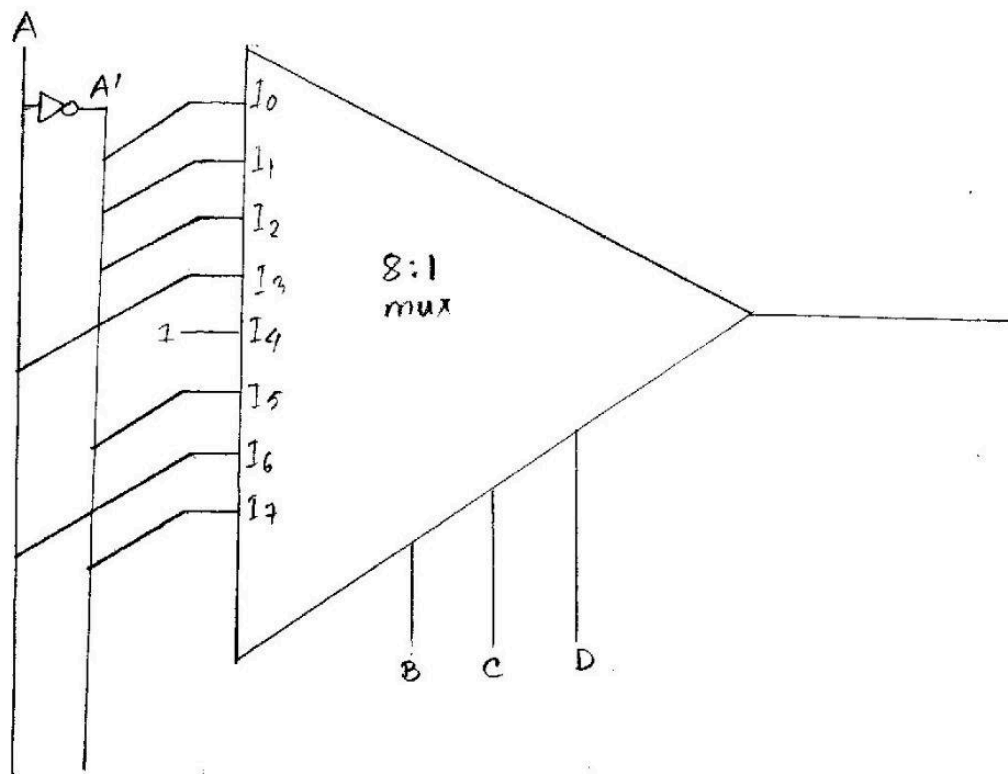


Ans to the ques no-04.

(a)

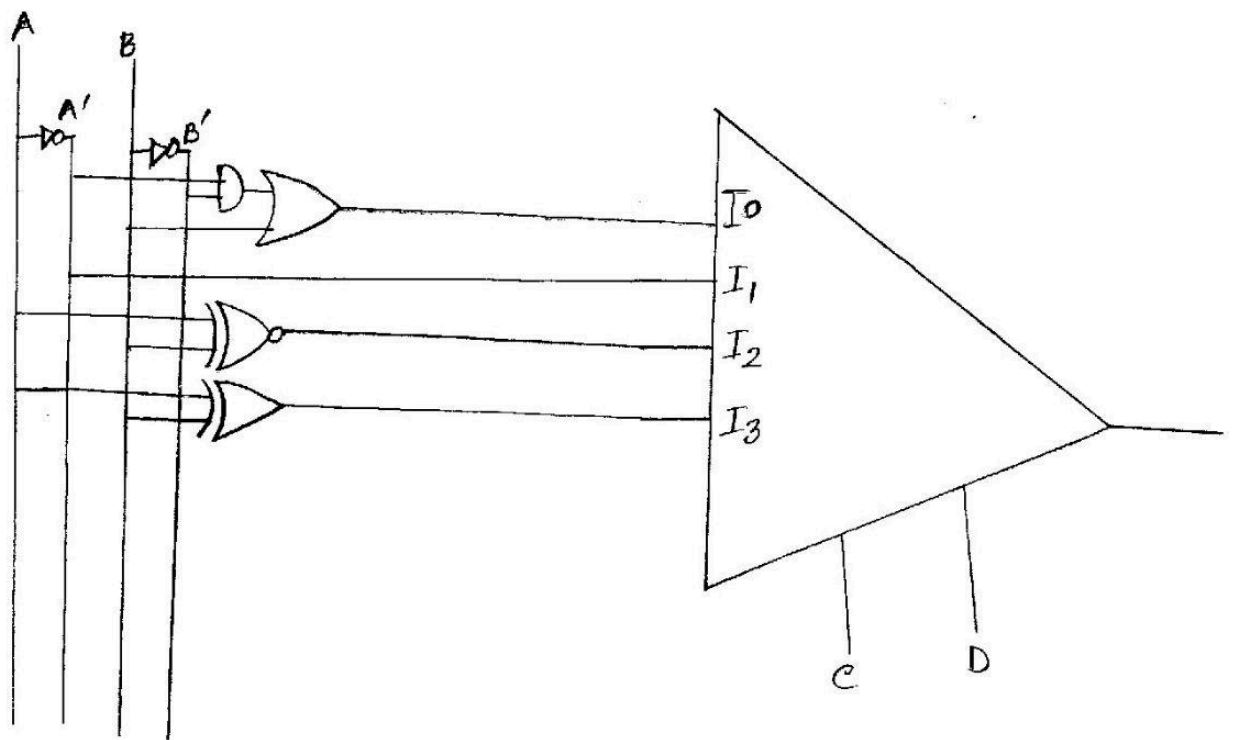


(b)



	I ₀	I ₁	I ₂	I ₃	I ₄	I ₅	I ₆	I ₇
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	A'	A'	A'	A	1	A'	A	A'

(c)



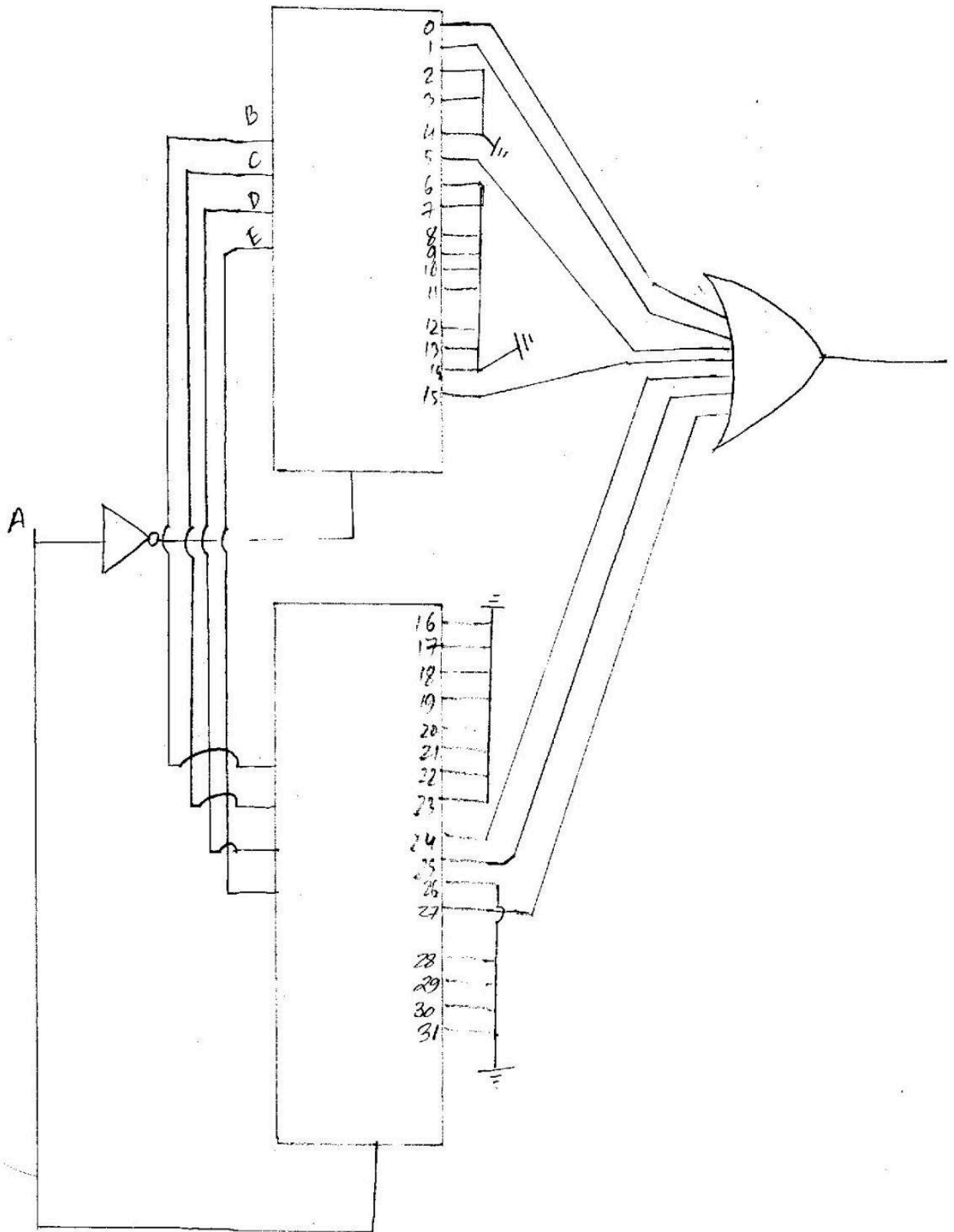
	I_0	I_1	I_2	I_3
$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15

$$\begin{aligned} I_0 &= A'B' + A'B + AB \\ &= A'B' + B(A' + A) \\ &= A'B' + B \end{aligned} \quad \left. \begin{aligned} I_1 &= A'B' + A'B \\ &= A'(B + B') \\ &= A' \end{aligned} \right\}$$

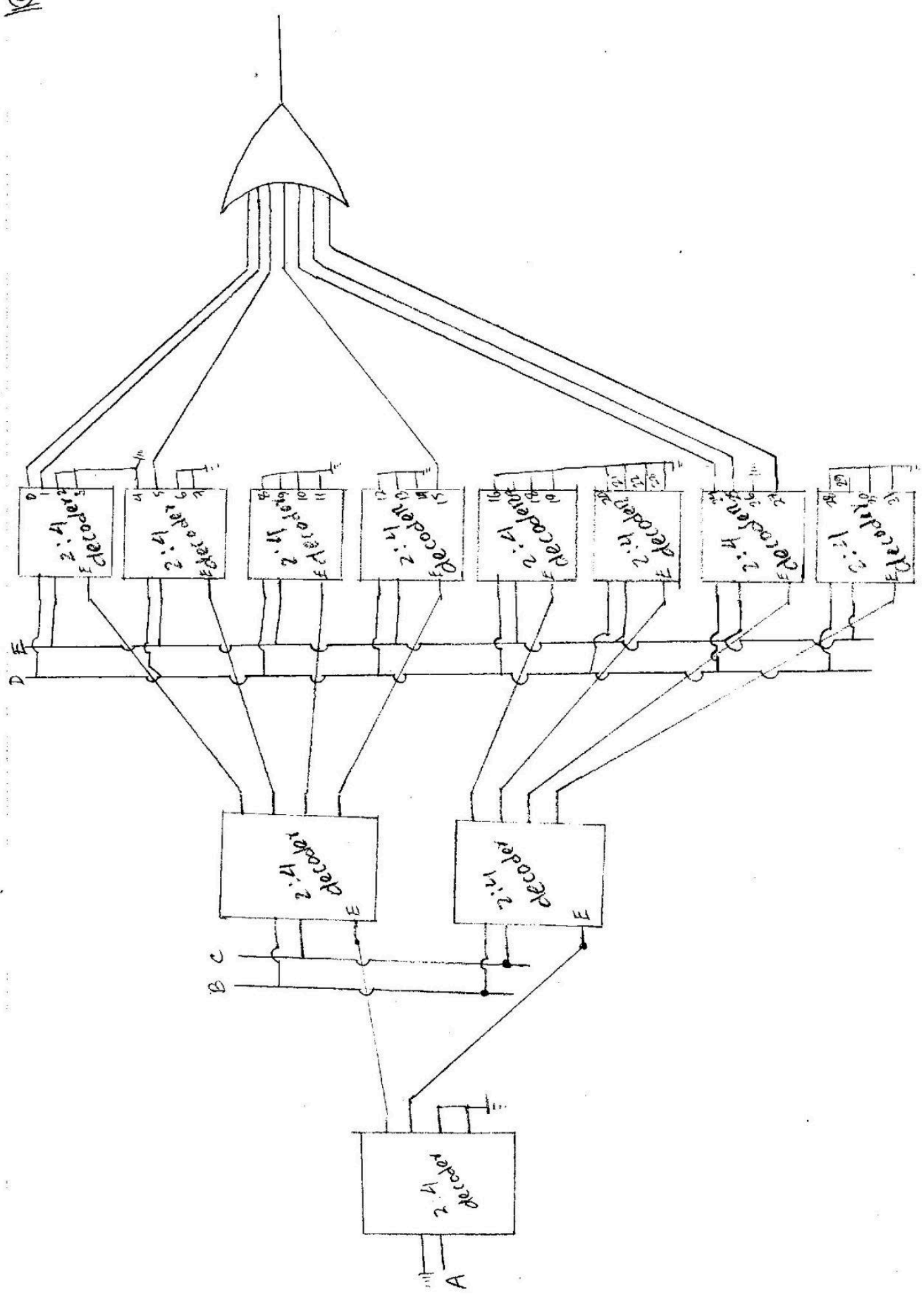
$$\begin{aligned} I_2 &= A'B' + AB \\ &= A \odot B \end{aligned} \quad \left. \begin{aligned} I_3 &= A'B + AB' \\ &= A \oplus B \end{aligned} \right\}$$

Ans to the ques no-05

(a)

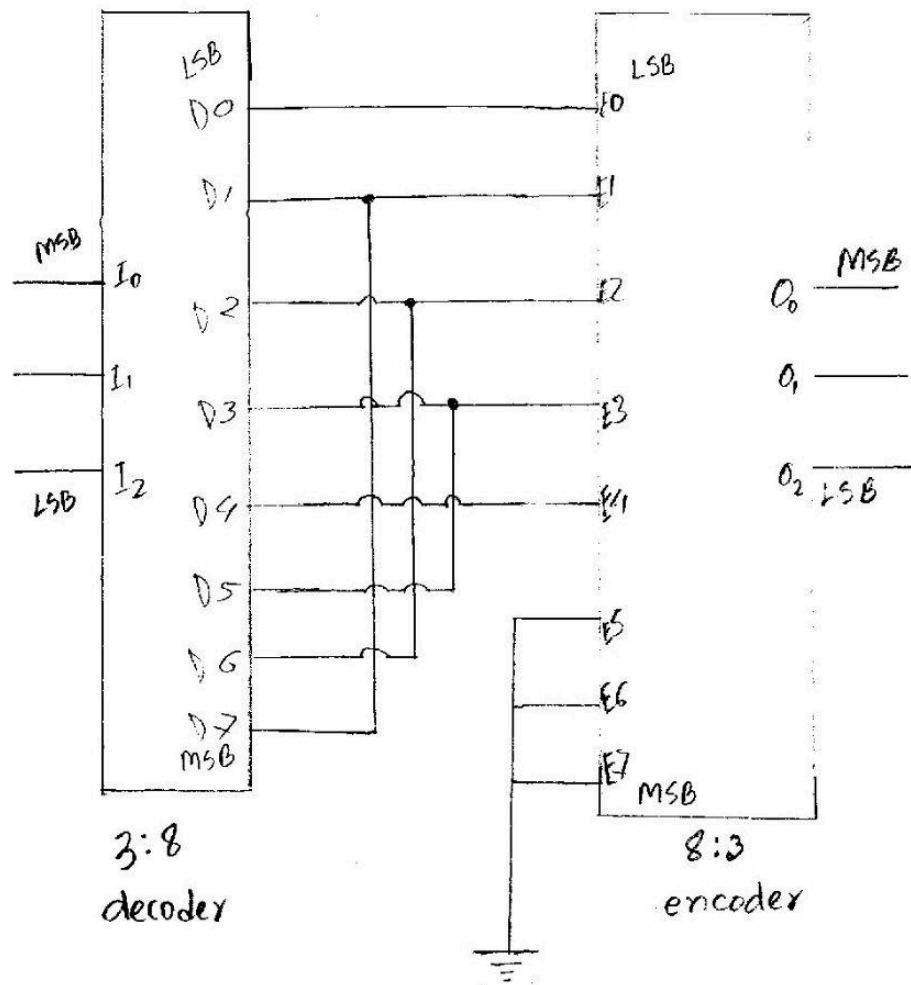


(b)



Ans to the ques no-02

The circuit that converts a 3 bit 2's complement number to its actual 3 bit binary number using encoder(s) and decoder(s). only.



Input	Decimal	Magnitude-output	input	Decimal	Magnitude output
000	0	000 D_0	101	-3	011 D_5
001	1	001 D_1	110	-2	010 D_6
010	2	010 D_2	111	-1	001 D_7
011	3	011 D_3	∴ Here, D_i = 'output of decoder, E_i = input of encoder $D_0 \rightarrow E_0$ $D_1 + D_7 \rightarrow E_1$ $D_3 + D_5 \rightarrow E_3$ $D_4 \rightarrow E_4$ $D_2 + D_6 \rightarrow E_2$		
100	-4	100 D_4			